Close to the cap: who plays the whole match across nine sports

Abstract

Governing bodies keep changing how many substitutions a team may make. We show who plays the whole match in nine sports and, for the first time across sports, how much position matters. We analyse each sport's rulebook and over a million appearances. Coaches choose which starters come off. But the cap on substitutions limits how many can. So before a rule change, governing bodies can compute the lowest share of starters playing the whole match. Teams come close to that share. When football moved from three substitutions to five, the share of outfield starters playing the whole match fell from 71 to 56 per cent. But football's goalkeepers stayed at 99 per cent. Position mattered more than the cap. Counting changes against starters' minutes gives a free check on match records. This check flags 8 to 27 per cent of teams' records in nine leagues of one public football dataset. In each flagged record, a starter taken off at minute 90 or later is credited with the whole match. For match simulators, we built a model of the pauses in which teams make their changes. It met all six predictions we wrote down before testing it on unseen matches.

Keywords: substitution rules; sports rules; match records; order statistics; underdispersion; data quality.

1 Introduction

Most team sports let a team replace players during a match, and most of them limit how many. We call the number of substitutions the rules allow the *substitution limit*, or the cap for short. Governing bodies keep changing the cap, for player welfare, for the flow of the match and for fairness between teams. Football, for example, moved from three substitutions to five (The International Football Association Board, 2020, 2022). Rugby union lengthened its bench to eight replacements. Gaelic football and hurling, the two Irish codes, allow five substitutions. Baseball sets no cap, but a replaced player may never return. In all of these sports a replaced player cannot come back. The only exceptions are temporary replacements for blood or head injuries, and a few returns that some seasons allowed (Table 1). We call these sports *one-way sports*. Rugby league and Australian rules let a player leave and come back. These changes are called interchanges, and both sports cap them. Rugby league has cut its cap to eight interchanges, and Australian rules has cut its cap twice and now allows 75. Ice hockey and basketball let players come and go without any cap. We call these four *re-entry sports*.

Each change of cap is a decision with practical stakes. A governing body needs to know how many starters a new cap can leave on for the whole match, and which ones. A club needs to know which positions play the whole match under each cap, so that it can plan their load. An injury researcher needs to know whether the match records say who really played the whole match. So does anyone who learns from match data, from public datasets (Pappalardo et al., 2019) to a club's own (Wang et al., 2024). A match simulator needs to know how many changes teams make, and when. Examples are the in-play models that join goals and cards (Titman et al., 2015) or that model added time (Maia et al., 2026). This paper answers these questions from the rules

and match records of nine sports: football, rugby union, rugby league, Australian rules, Gaelic football, hurling, baseball, basketball and ice hockey.

A substitution limit works like a speed limit. Such a limit does not set how fast a driver goes, but it caps how far a car can travel in an hour. In the same way, a cap does not choose which players a coach takes off, but it caps how many starters a team can take off. Teams drive close to their cap. Under five substitutions, for example, football's teams take off 39 per cent of their starters by substitution, and the cap allows at most 45.5 per cent. Under three substitutions they took off 26 per cent, against a ceiling of 27.3. In Australian rules, which cut its cap twice, teams used 91 to 95 per cent of each cap.

Where a replaced player cannot return, simple counting links the starters who leave to the changes a team makes. Each change takes off either a starter or a player who came on earlier. So the starters who leave before the end equal the team's changes, with two kinds of correction. Changes that take off an earlier substitute are subtracted, and starters sent off or otherwise not replaced are added. An equation that holds by counting alone, like this one, is called an *identity*. Where no player may return even for a time, a record's changes and its players' minutes must agree, so a record where they disagree is inconsistent. That gives governing bodies a formula and data providers a free check on match records.

The cost of ignoring the counting.

- *Credit given for the rule.* An analysis that credits coaches with how many starters they take off also credits them with a ceiling the rule sets.

- *Caps that protect fewer players than they seem to.* A cap raised to spread the load still leaves at least 54.5 per cent of football's starters playing the whole match under five substitutions, less any sent off. Those players may now face higher demands (López-Valenciano et al., 2023), because substitutes run harder (Allen et al., 2025).

- *Records that misreport who finished.* A record that lists a late change at minute 90 credits its starter with the whole match. In a large public football dataset, up to 27 per cent of the teams' records carry such an entry. This matters, because rule changes for player safety have been judged on public media records (Beaudouin et al., 2019).

- *Simulated matches that run wrong.* A simulator that picks the minute of each change at random, one by one, puts two consecutive changes in one minute 6 per cent of the time. The records show 37 per cent.

What this means in practice. The results answer the four questions.

- *For governing bodies.* From the cap alone, a governing body can compute the most starters a new cap lets a team take off. It can do so before any rule change. Teams come close to that maximum, because they use most of their cap and coaches rarely replace an earlier substitute. Under each cap of ten or fewer measured here, about half or more of the starters play the whole match, even where players may come back on. In rugby league most starters still play

the whole match under its eight interchanges. In football the starters taken off and the changes made differ by less than 3 per cent of the team, in each month of the season.

- *For clubs.* In football and rugby league, who plays the whole match depends more on position than on the cap. Football's goalkeepers, rugby league's backs and baseball's starting batters finish far more often than outfield starters, forwards and starting pitchers. So their load is managed between matches, not by substitution.

- *For injury research and data providers.* Where a replaced player cannot return, a record's list of changes and its players' minutes must tell the same story. That simple check flags 8 to 27 per cent of the teams' records in nine football leagues. In each flagged record, a starter taken off at minute 90 or later is credited with the whole match. On average the rugby league and Australian rules records pass a second check: their minutes sum to the players on the field times the length of the match.

- *For simulators.* We summarize the number of changes a team makes with a curve cut off at the cap. Its width is about one to two changes, and its centre moves with the limit. A team's changes come, on average, near where the same number of random moments in one stretch of the match would fall. A model of the stoppages, the pauses in which a team makes its changes, met all six pass marks. We wrote them down before we read the matches of 2020 that we kept back to test it.

What is new. Earlier work has studied substitutions for two decades, mostly as a coaching decision. It has modelled the best moments to change players (Hirotsu & Wright, 2002, 2005; Hirotsu et al., 2024), timed the changes (Del Corral et al., 2008; Myers, 2012; Silva & Swartz,

2016; Hills et al., 2019; Sun et al., 2024) and related them to the score (Gómez et al., 2016; Wittkugel et al., 2022). It has measured how their use grew with each change of rule (Nevill et al., 2026; Dilger & Vischer, 2024), how often teams use their whole limit (Ribeiro et al., 2020; Meyer & Klatt, 2021; Roberts et al., 2024; World Rugby, 2020) and how often they make two changes at once (Xiao & Zhang, 2024). It has compared rugby league's whole-match and rotated players (Black & Gabbett, 2014; LeagueEyeTest, 2021), and the limits of different sports as rates (Mota et al., 2020). None of it asks what the rule itself caps, such as the share of starters taken off. This paper studies a rule already in force, not a proposal for a new one (Wright, 2014).

We know of no earlier work, apart from the preprint described below, that does any of five things this paper does. It compares who plays the whole match position by position across sports. It checks match records against the identity. It counts how often a substitute is replaced in turn, change by change. It models how many changes teams make and how they group them into stoppages. And it tests a model of the changes' minutes on matches the model had not seen. Ricou (2026) first reported, sport by sport, how many starters play the whole match and how closely the starters taken off track the substitutions made.

2 Rules and records

Table 1 gives each sport's limit and the records we read. The Electronic Companion, the paper's online appendix, lists the key terms the paper uses. We write n_f for the players a team starts with, and $n_{\max}$ for the players a team may bring on. In a re-entry sport the rule limits interchange events rather than players, and we write that event limit n_e. Some matches allow extra replacements, so we also use the *effective limit* n_{eff}, the limit that actually applies in that match. Since 2020 each football competition may choose three or five substitutions (The

International Football Association Board, 2020, 2022), so football's limit is read from each competition's own lineups.

Every record comes from public match data. Football's events and appearances come from a public dataset built from Transfermarkt's match pages (Cariboo, 2026). The other sports' records come from the leagues' own pages and statistics interfaces (MLB Advanced Media, 2026; National Hockey League, 2026; Australian Football League, 2026; AFL Tables, 2026; National Rugby League, 2026; It's Rugby, 2026; Euroleague Basketball, 2026; Gaelic Athletic Association, 2026). We call the records read for one analysis its *panel*. We read each panel one team-match at a time. A *team-match* is one team's record of one match. The Electronic Companion lists the panel behind each result. The three-substitution panels leave out the matches played under the temporary five-substitution amendment of 8 May to 31 July 2020. The count and timing panels leave out the 2019–20 season altogether. Its matches under the amendment are kept apart to test the model of Section 5. We call them the *held-out matches*.

In the re-entry sports, the minutes t_i of a team's players should add up to the players on the field times the length of the match t_M, so that $\sum_i t_i = n_f\, t_M$. Injury researchers call this total *exposure* (Fuller et al., 2006). Each of 10 panels in two sports passes this check on average (Electronic Companion). In the Australian Football League (AFL), the top competition of Australian rules, the sum is within 0.03 per cent of that total. But the league's own interchange records may make it so automatically.

Table 1: The rules and the records. The limit is $n_{\max}$ in the one-way sports and the event limit n_e in rugby league and Australian rules. 'Not counted' lists the replacements the rule leaves outside the limit. The Electronic Companion gives each rule with its reference and explains season names.

Sport	n_f	Limit	Not counted	Seasons	Panel and source	Team-matches
Association football	11	3; 5 from 2020 by the competition's choice, in at most three stoppages in play; +1 in extra time	concussion from 2020	2012–25	29 men's leagues, Transfermarkt; event records name the player replaced in nine	59,982 (five); 58,328 (three); 42,303 (identity)
women				2024	seven leagues, Soccerdonna and FBref; the month-by-month identity only	1,996
Rugby union	15	7, then 8 by league from 2008; 8 in the Laws from 2014; up to four returns in the French leagues, 2018–25	blood, head injury	2004–26	Five leagues, itsrugby.fr; the identity also on the 2025–26 Top 14 sheets	25,028 (identity); 25,852 (count)
Gaelic football	15	5 to 2013, 6 in 2014–19, 5 from 2020; +3 in extra time	blood, head injury	2002–24	All-Ireland senior finals, match reports	10
Hurling	15	5; +3 in extra time	blood, head injury	2003–25	All-Ireland senior finals, match reports	18
Baseball	10	none; at most 13 pitchers from 2022; a pitcher faces three batters from 2020	–	2008–25	MLB Stats API game logs; the identity on 2024	77,096 (count); 4,858 (2024)
Rugby league	13	10 to 2015, 8 from 2016; bench 4	head injury, 18th player	2013–25	NRL match-centre feed	5,061
Australian rules	18	none to 2013, 120 in 2014–15, 90 in 2016–20, 75 from 2021; bench 4, 5	blood, head injury, changes at the breaks, the substitute	2013–25	AFL Tables and the AFL's statistics API, men and women	2,916 (men); 198 (women)
Ice hockey	6	none; 20 dressed	–	2008–24	NHL game sheets and shift charts, public statistics API	2,080 (2023); 730,501 skater-games
Basketball	5	none; 12 dressed	–	2007–24	EuroLeague and EuroCup box scores, 29 competition-seasons	10,830

3 How many starters leave

3.1 Who plays the whole match

We say that a starter plays the whole match when the starter is on the field from the start to the final whistle. In football, 73 per cent of the starters played the whole match under three substitutions and 60 per cent under five. In rugby union, with a bench of eight, about half the starters do. The share is 52.3 per cent on the lineups up to 2024–25 and 49.9 per cent on the 2025–26 lineups. We also read the All-Ireland senior finals of Gaelic football and hurling, keeping those with complete and consistent lineups. In them the shares are 75 and 71 per cent, over five and nine finals. In the National Rugby League (NRL), the top rugby league competition of Australia and New Zealand, a player may come back on. Yet 63 per cent of its starters play the whole match under eight interchanges among 13 players, more than in football under five substitutions. Under a cap several times the size of the team, as in Australian rules, almost everyone comes off. Only 3 per cent of its appearances last the whole match under a cap of 75 interchanges, and 1 per cent under a cap of 90.

In football and rugby league, position matters more than the cap. In rugby league, for example, the backs last the whole match in 91 per cent of their appearances under eight interchanges. The forwards, whom the interchanges rotate, last it in 31 per cent. The cut from ten interchanges to eight moved neither group by more than 1.1 percentage points, far less than the difference between the two groups. In football the goalkeeper finishes 99 per cent of matches under five substitutions, as under three. An outfield starter finishes 56 per cent, down from 71 under three. In baseball the starting batters stay in the game to its end in 84 to 88 per cent of games. The starting pitcher does so in only 0.6 per cent, because relief pitchers take over during the game. Where

players change on the fly, as in ice hockey and basketball, almost no one plays the whole match.

3.2 The counting behind it

Counting explains why so many starters finish where no one can return. In a one-way sport each substitution takes one player off, and that player cannot return unless the change was a temporary replacement. A team makes only as many substitutions as its cap allows. The player taken off is either a starter or a substitute who came on earlier. We call the second kind of substitution a *chain*. So every substitution that is not a chain takes off one starter. A starter recorded as leaving before the end of the match is *withdrawn*, and a team's *withdrawal rate* s_w is the share of its starters withdrawn.

Proposition 1. *In a one-way sport, let a team-match have S substitutions, of which C are chains, L take off a starter whom the record credits with the full match, and X take off a player on neither the starters' nor the substitutes' list. Let D be the starters sent off and not replaced, and U the other starters recorded short with no substitution or card. Then*

$$n_f \left(\frac{S}{n_f} - s_w \right) \;=\; C + L + X - D - U, \tag{1}$$

and, because $S \le n_{\text{eff}}$,

$$s_w \;\le\; \frac{n_{\text{eff}} + D + U}{n_f}. \tag{2}$$

The proof is only counting, and the Electronic Companion writes it out line by line. In words, a team's substitution rate S/n_f, its substitutions per player on the field, equals its withdrawal rate apart from five corrections. Of the five, only chains are a coach's choice. A

late entry is a change made at minute 90 or later, in the added time at the end of the match. The record lists it at minute 90, so its starter seems to finish. Records of goals spike at minutes 45 and 90 for the same reason (Forrest & McHale, 2019). Late entries and players on neither list come from the record, not the match. Starters sent off and other exits without a replacement are *departures without replacement*. They are events of the match, and the record gets them right. Take a football team that makes five substitutions. One of them takes off a player who had come on earlier, so it is a chain. Another comes in added time, and the record lists its starter as finishing the match. The record then shows three of its 11 starters withdrawn, not five. Each of the two missing starters is one of the five corrections. The second equation is the bound the cap sets. Its substitution part, n_{eff}/n_f, needs no data. A football team allowed five substitutions, for example, can take off by substitution at most five of its eleven starters, 45.5 per cent, whatever its coach decides.

3.3 Almost every change takes off a starter

Football's event records name, for every substitution, the player who left and the player who came on. So in nine leagues every part of equation (1) is counted, on 42,303 team-matches. We call the share of substitutions that are chains the *chain share* $s_c = C/S$. In these leagues it is only 0.2 to 0.9 per cent. We call one league under one substitution rule a *league-regime*. Figure 1 places every league-regime with the substitution rate across and the withdrawal rate up. We call the difference between the two rates the *gap*, and give it in per cent of the team. Every one of football's 18 league-regimes has a gap below 3 per cent, which is 0.33 of a starter per team-match. We call this level the *tolerance*.

Up to 2024–25 rugby union's lineups list each replacement with

the minute of the starter it replaced, so its two counts come from one list. On the lineups of five leagues, 25,028 team-matches give a gap of 0.2 per cent. This checks that the lineups are consistent, not how many chains there are. In the Gaelic finals the gap is exactly zero in every team-final but one in each code. The gap also stays small in each month of the season. Read from the event records, its largest monthly value in the men's European leagues is 2.3 per cent, inside the tolerance.

3.4 Checking the records

On records that name the player replaced, every part of the identity is counted. We flag a team's record when it has at least one late entry or one player on neither list. A flag needs only the record's list of changes and its players' minutes. It marks a starter taken off by a substitution yet credited with the full match, or a player taken off who is on neither list. In the records here 8 to 27 per cent of the teams' records are flagged. Every flag is a late entry. The source lists added time at minute 90, so the starter seems to finish. Figure 2a compares each league's flags under three and five substitutions, and per team-match they rose in every league. Teams also made more changes. Per substitution, the late entries rose in seven of the nine leagues, stayed level in Italy and fell in the Netherlands. Counted on the records, the gap is mostly these late entries: chains are a sliver, and departures without replacement pull it the other way (Figure 2b).

4 How many changes a team makes

The number of changes varies little. A common measure of spread is the standard deviation, the typical distance from the average. For the number of changes a team makes, it is 0.44 to 1.25

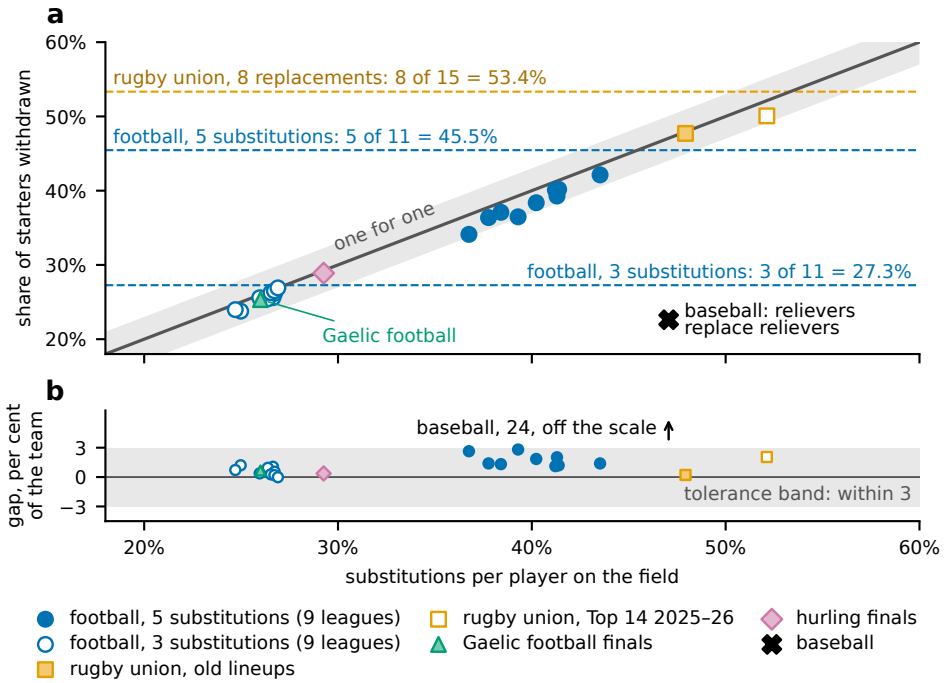


Figure 1: Substitutions take off starters almost one for one. (a) Each point is one league under one substitution rule, a league-regime, with its substitutions per player on the field across and the share of its starters withdrawn up. The diagonal is one starter withdrawn per substitution, and the grey tolerance band spans 3 per cent of the team either side of it. Football's filled points are under five substitutions and its open points under three, and each dashed line is the ceiling a cap sets, its share rounded up. Rugby union's old lineups and the Gaelic finals give both counts from one list, so their agreement is close to built in (lighter markers). The open square is France's Top 14 in 2025–26, on the newest lineups, which list every player's minutes. Baseball lies far below the band, because at least 45 per cent of its substitutions take off a player who had come on earlier, a chain. (b) Each gap, the substitution rate less the withdrawal rate, in per cent of the team, with the same band. Figure EC1 draws each gap with its 95 per cent interval, which shows how precise it is.

Alt text: Panel a: a scatter with a diagonal line inside a grey band. The eighteen football points lie in the band below their dashed cap lines, with the rugby union, Gaelic football and hurling points, and one baseball point sits far below it. Panel b: the same points as gaps near zero inside a narrow band, with an arrow for baseball off the top.

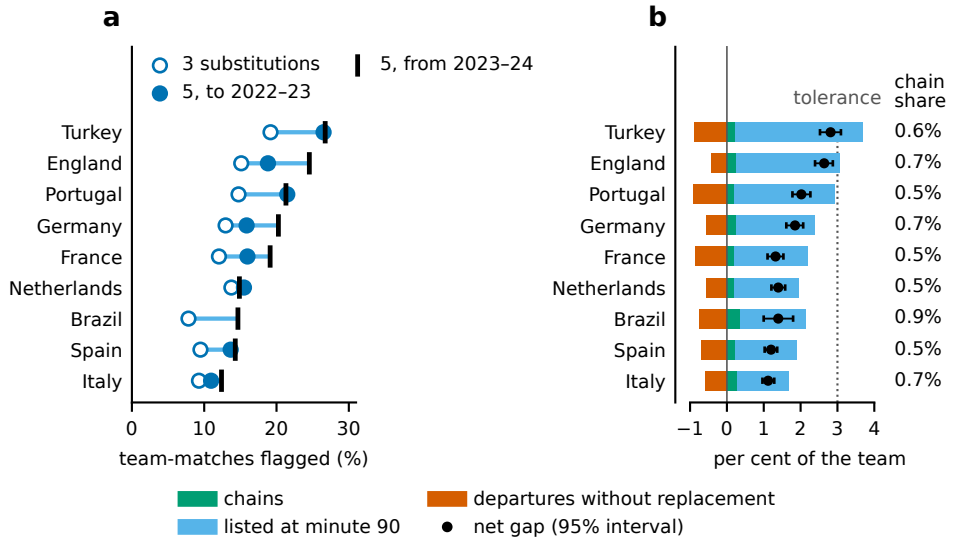


Figure 2: Counting flags 8 to 27 per cent of football's team-matches, every flag a change listed at minute 90. Per team-match the flags rose in every league at the move to five substitutions, and in England and Germany most of the rise came from 2023–24. In England that was the season the Premier League began to add the exact time lost. (a) The share of team-matches flagged in each league, pooled over its seasons under three substitutions (open), under five before 2023–24 (filled) and under five from 2023–24 (black tick), joined in that order; a team-match is flagged when its record credits a starter taken off by a substitution with the whole match. Brazil's only season under five substitutions is 2024, so it has a tick and no filled circle. The two rules' seasons are different years. (b) Football's gap under five substitutions, part by part, in per cent of the team and in the rows of (a): chains, the substitutions that take off a player who had come on earlier; changes listed at minute 90; and, to the left, departures without replacement, starters sent off or leaving unreplaced. By Proposition 1 the parts sum to the gap exactly. The black mark is the net gap with its 95 per cent interval, the dotted line its tolerance, and the right column the chain share, the chains' share of the substitutions.

Alt text: Panel a: for nine football leagues, a line runs from an open circle through a filled circle to a black tick; every filled circle lies right of its open circle, and the tick lies furthest right of the filled circle in England and Germany and close to it in the Netherlands, Portugal and Turkey. Panel b: for the same leagues, horizontal bars made mostly of changes listed at minute 90, with a thin green sliver of chains, a shorter bar to the left for departures without replacement, a black dot for each net gap and a dotted line for the tolerance to the right of every dot.

changes. The standard model of random match events is a Poisson count (Maher, 1982; Dixon & Coles, 1997). In football it puts 11.1 per cent of teams in the wrong count (Figure 3). A narrow curve cut off at the cap, set out below, puts 0.6 per cent there, and at most 4.2 per cent in the other count panels.

Narrower than Poisson, wider than a binomial. A Poisson count's variance, the square of its standard deviation, equals its mean, or average. The number of changes is far narrower. In the five count panels we read, its variance is only 0.07 to 0.52 of its mean. A limit close to the average leaves the count little room to vary (Bhatia & Davis, 2000). This explains part of the narrowness. The usual model for a count with a ceiling is a binomial count on the cap. It treats each allowed change as a separate yes-or-no choice, made with the same chance. In football and rugby union the count is wider than that binomial, with 1.4 and 1.8 times its variance. The reason is that many teams use the whole cap and some use only a few changes. Basketball's count is about as narrow as a binomial, and baseball's, with no ceiling near it, is narrower than Poisson (Electronic Companion). For a governing body the telling number is the share of teams at the cap: 85 per cent under three substitutions, 56 per cent under five and 56 per cent in rugby union.

A narrow curve, cut off at the limit. We summarize the count with a normal distribution, the bell-shaped curve, with centre μ and standard deviation σ. We round it to whole changes and cut it off at the limit. We call this curve the *censored normal*. For each sport we fit it, choosing the centre and width that match the data best. Its centre and width describe the curve, not any one team's plan. Its width, its standard deviation before the cap cuts it, is 0.95 to 1.76 changes in every count panel. That is more than the observed standard deviation

wherever the cap cuts off the top. Its centre moves with the limit. In rugby union the centre is 2.20 replacements higher in the seasons with a bench of eight than in the earlier seasons with seven. It can move by more than one replacement, because it is the centre of the curve before the cut.

5 When the changes come

Timing read from the count. A simple line, built below from the number of changes, gives the average minute of each change (Figure 4a). It misses by 1.1 minutes in football and 1.3 in rugby union. Putting every change at the average change's minute would miss by 8.7 and 7.8 (football in Figure 4c).

A simple line for the timing. Consider the teams that make exactly n changes in a match. Take the k-th of those changes and average its minute across the teams. We call that average a *cell*. Now picture n moments dropped at random on a stretch of the match and put in order, which statisticians call order statistics. On average the k-th of them falls at the fraction $k/(n+1)$ of the stretch, a classical result of probability (Cunnane, 1978; David & Nagaraja, 2003). A team that makes three changes, for example, would put the first a quarter of the way through the stretch, the second halfway and the third three quarters of the way. We call the stretch of the match in which a team's changes fall its *window*. If a team's changes behave like such random moments on a window, the cells fall on a line,

$$\mathrm{E}[t_{k:n}] \;=\; t_M \left(\zeta_{0,n} + \zeta_1 \, \frac{k}{n+1} \right), \tag{3}$$

where $\zeta_{0,n}$ is where the window opens for a team with n changes and ζ_1 how wide it is, both as shares of the match. We call this the order-

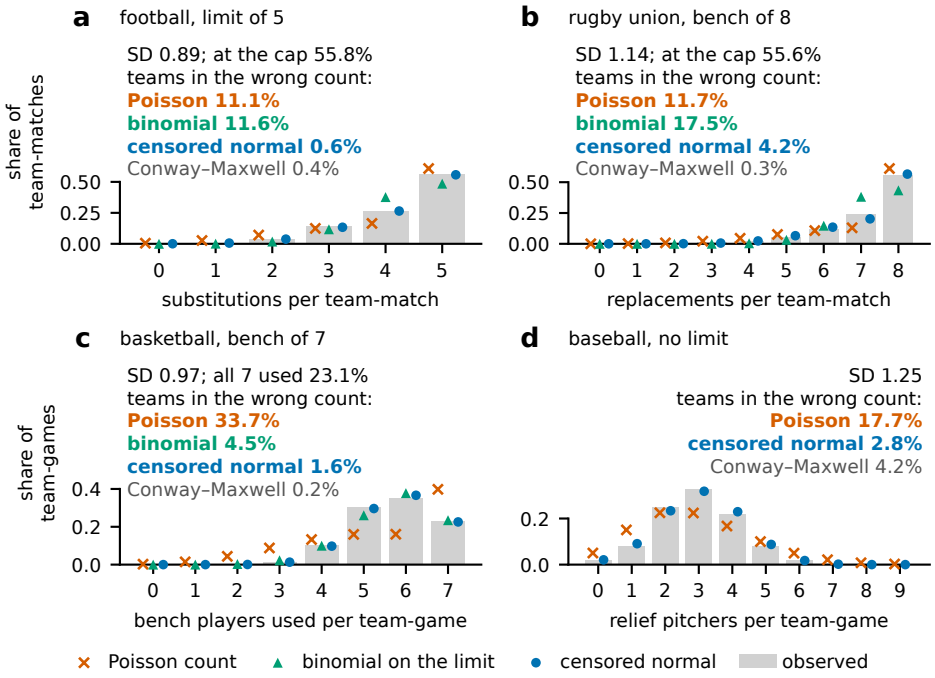


Figure 3: The number of changes varies little: the censored normal comes close to its shares, and a Poisson count does not. Grey bars are the observed shares, crosses the best-fitting Poisson count, triangles the binomial count on the limit with the observed mean (none in d), and dots the censored normal, each cut off at the limit. Each panel prints the observed standard deviation (SD), the share of teams at the cap (in basketball, the share that use all seven bench players), and the share of teams each model puts in the wrong count, that is, the share it would have to move to match the observed shares; its last line is the best Conway–Maxwell family, a standard model for counts narrower than Poisson, which is not drawn and which Figure EC3 compares. The Poisson count spreads teams too widely, and in football and rugby union the binomial is too narrow, because many teams use the whole cap and some use only a few changes. (a) Football under five substitutions. (b) Rugby union's sides with a bench of eight. (c) Basketball's bench players used, of the seven on its bench. (d) Baseball's relief pitchers, where no limit binds, shown to nine.

Alt text: Four bar charts of the number of changes per team. Blue dots sit on top of the grey bars. Red crosses miss them, too high at low counts and at the limit. Green triangles pile too much weight one change short of the limit in football and rugby union.

statistic line.

The line summarizes the averages. Figure 4a shows every cell of football's five-substitution era and of rugby union, and the line passes close to almost every cell. We estimate the line on the middle changes, with an opening for each number of changes. Football's window is then 43.7 minutes wide. Teams that make more changes start earlier: their window opens at 51.3 minutes for three, 50.8 for four and 48.5 for five.

Stoppages, and the bundling they require. The line gives the average minute of each change, but it cannot give a simulator the changes of one match. Minutes picked at random, one by one, put two consecutive changes in one minute 6 per cent of the time. In the records 37 per cent share a minute. The reason is that a team using five substitutes may make them in at most three pauses during play. We call each pause in which a team makes one or more changes a *stoppage*.

Proposition 2. *Let a team stop play for changes at most w times during play, besides half-time, and let it make n changes, h of them at half-time. Then at least* $\max(0,\, h-1)+\max(0,\, n-h-w)$ *of its* $n-1$ *consecutive pairs of changes share a stoppage.*

Half-time holds its h changes in one stoppage, and the $n-h$ changes made during play fall in at most w stoppages. So the fewer stoppages a team may use, the more of its changes must share one. With w at three and each team's own h, football's rule forces 35.0 per cent of consecutive changes into one stoppage.

A model of the stoppages. We built a model of the stoppages in three steps from football's five-substitution seasons, which we call the *fitted seasons*. A stoppage here is the changes listed in one minute.

First, a team's number of changes is picked at random from the censored normal of Section 4. Second, the team splits its changes between half-time and its stoppages in play, in the proportions that teams with the same number of changes used. Third, a team with m stoppages in play picks their minutes at random from the minutes at which teams with m stoppages in play stopped play. It picks again if two minutes are the same, and puts them in order. So its j-th stoppage in play comes at the j-th of m ordered random minutes,

$$\begin{aligned} &\tau_{j:m}, \qquad \tau_1, \ldots, \tau_m \text{ drawn independently from } F_m, \\ &\text{a set with a repeated minute being redrawn,} \end{aligned} \tag{4}$$

where F_m is the distribution of the minutes at which teams with m stoppages in play stopped play. Each change then takes its stoppage's minute. The order-statistic line of equation (3) is the case in which every change is its own stoppage and F_m is spread evenly over the window. On the fitted seasons the model misses the cells' average minutes by 0.6 minutes, against 1.1 for the line. The model is built to match the share of changes made in one minute.

We then tested the model on the held-out matches, which no part of this paper had read. They are the 1,270 team-matches of the 2019–20 season played under the temporary amendment of May 2020. We wrote down six predictions with their pass marks in a register before reading them, and every one held (Electronic Companion). The model missed the cells' average minutes by 0.9 minutes, against 1.5 for the line. It put 37 per cent of consecutive changes in one minute, against 33 per cent in these matches' records (Figure 4b). After the test we also described the held-out matches in other ways (Electronic Companion). The later fitted seasons' own cell averages, carried back to these matches, missed by 0.7 minutes. That is slightly less than the model. But cell averages give a simulator no sequences of changes, and the model does.

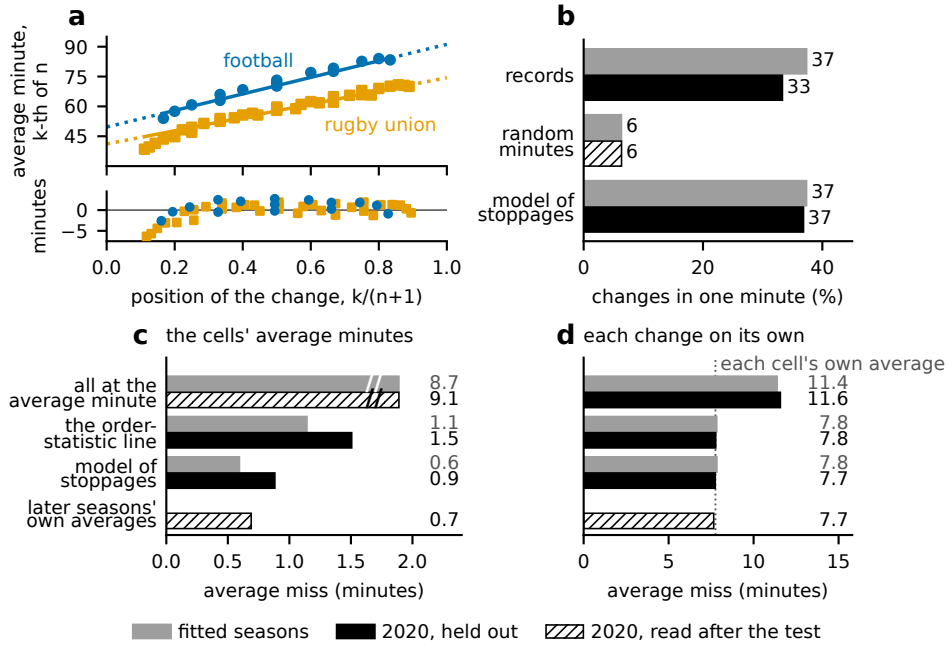


Figure 4: The average minute of each change can be read from the number of changes, and changes come bundled in stoppages. (a) Each point, a cell, is the average minute of the k-th of n changes in football's five-substitution era and in rugby union, with the order-statistic line, drawn here with one opening shared by all n, dotted past the cells; the strip gives each cell's distance from it in minutes. First changes come early, partly for injuries. Rugby union's cells sit lower because its match lasts 80 minutes. (b) Consecutive changes made in one minute: in the records, for minutes picked at random and for the model of stoppages, which is built to match the fitted seasons. (c) How far each placement misses football's cell averages, each cell weighted by its changes, on the fitted seasons' 15 cells and the 14 held-out cells with at least 50 changes; the average minute's bars are cut. The later seasons' own averages, with no grey bar as the fitted seasons are their source, miss slightly less than the model, a difference whose 95 per cent interval includes zero. (d) The same for each change on its own; the dotted line is the miss of each cell's own average on the fitted seasons, 7.8 minutes. In (b) and (c) the black bars answer predictions registered before the held-out matches were read; hatched bars were read after the test.

Alt text: Panel a: points along two rising lines, football above rugby union, most within two minutes of their line. Panels b to d: bars, short in b for random minutes, cut in c for the average minute, near eight minutes in d.

6 Discussion

6.1 Managerial implications

For governing bodies. The counting gives governing bodies a simple formula. A limit of n_{eff} caps the share of starters a team can withdraw by substitution at n_{eff}/n_f. So it guarantees that at least $1 - n_{\text{eff}}/n_f$ of them finish, less departures without replacement. We call this share the *floor*. Figure 5a turns this formula into a design chart, and Figure 5b sets the spread between positions beside each cap. A sixth substitution, for example, would raise football's ceiling from 45.5 to 54.6 per cent of the team. At least 45.4 per cent of the starters would still finish. The Gaelic Athletic Association's football advisory group has proposed a sixth substitute (Harrington, 2026). It would lower Gaelic football's floor from two thirds of the starters to 60 per cent. The floor is useful because teams came close to the caps they had. The share of football's starters who finished sat 0.4 percentage points above the floor under three substitutions and 5.3 under five. Rugby union's sat 3.3 to 5.6 points above it. How close teams would come to a new cap is a forecast. When football moved to five substitutions, teams used a smaller share of the larger cap. Proposition 2 does the same for stoppages: it tells a governing body how many changes any limit on stoppages makes teams combine. Keeping today's numbers of changes and half-time changes, two stoppages in play would force at least 62 per cent of consecutive changes into one stoppage. Four would force at least 14 per cent.

What else weighs on the decision. The formula is one input among several.

- *Player welfare.* Load, including a congested calendar, is a major risk factor for injury (Soligard et al., 2016). In the Premier League, five substitutions seem to have given clubs more room to manage their squads (Allen et al., 2025).

- *Head injuries.* Former professional footballers die of brain diseases such as dementia at several times the rate of similar men (Mackay et al., 2019), most of all outfield players (Ueda et al., 2023; Russell et al., 2021). Every rulebook here that sets a limit now allows replacements for head injuries outside it, so the formula shows how many more starters each exemption lets a team take off.

- *Tactics and results.* Substitutes cover more high-intensity running than the players they replace (Bradley et al., 2014; Chen et al., 2025), so a larger limit gives a coach more of this lever.

- *Fairness.* Teams whose players are available more often finish higher (Hägglund et al., 2013). A larger limit lets a deep squad bring on more of its quality than a thin one can.

- *The flow of the match.* Every stoppage for changes holds up play, which is why football's five substitutions must come in at most three stoppages during play (The International Football Association Board, 2020). From 2026–27 a substituted player has 10 seconds to leave the field (The International Football Association Board, 2026). The changes that take off a starter at minute 90 or later give a baseline for judging that rule (Electronic Companion).

- *Spectators and broadcasters.* Every stoppage for changes is one that spectators sit through and broadcasters must fill. No study we know of has measured how the limit changes their experience.

- *Heat.* In hot and humid matches players sprint less (Nassis et al., 2015), so a limit set for a cool league may matter more in a hot one.
- *Cost and young players.* A larger limit makes a deep squad more valuable (Allen et al., 2025). Each extra change also opens minutes that a club could give to a young player, which no study we know of has measured.

The arithmetic does not rank these goals. It tells a governing body exactly how far each option moves the share of starters who can be withdrawn.

For statistics units and data providers. In sports where no player may return even for a time, the identity gives a check at two levels. On records that name the player replaced, a data provider can flag each team-match with a late entry or a player on neither list. Where records do not pair players, as in most public appearance tables, the gap checks them across many matches once the starters sent off are added back. A source that lists added time at minute 90 should then show a clearly positive sum, 0.8 to 3.6 per cent of the team here. So a sum near zero points to changes missing from the record, as in a first reading of Brazil's top league (Electronic Companion). Such a source's minutes also stop at minute 90 for every player. So they leave out added time for all of them. The identity cannot see that shortfall, because every player shares it. So injury research must add the added time back to measure exposure. A provider that keeps added time in its minutes, as Wyscout does (Wyscout, 2026), raises no flag.

For builders of match simulators. A simulator can pick the number of changes at random from a censored normal. It can also use a Conway–Maxwell family, a standard model for counts narrower than Poisson (Shmueli et al., 2005; Kadane, 2016). It can then pick the

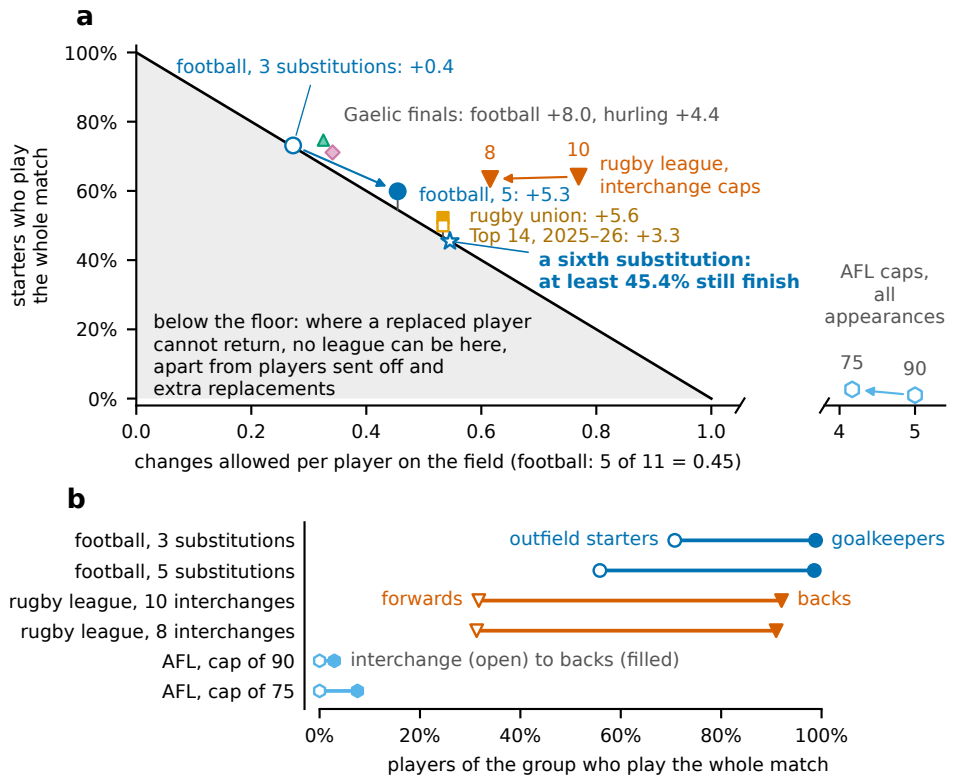


Figure 5: A cap sets a floor under the starters who finish, and the leagues where a replaced player cannot return sit close above it. (a) Where a replaced player cannot return, at least $1 - n_{\text{eff}}/n_f$ of the starters finish, less those sent off: the line is this floor, and the region below it is shaded. Labels and thin grey lines give each league's distance above it in percentage points; the Gaelic finals are plotted at five substitutes, though the 2015 football final allowed six. Football is read at the final whistle, and rugby union's open square is the Top 14 on its newest lineups. Arrows join each change of cap. Rugby league and the AFL let players return, so their caps count interchanges; the AFL sits in a compressed segment. (b) Under each cap, the group of positions that plays the whole match most often (filled) and least often (open). In football and rugby league the groups differ more than a change of cap moved them, and in the AFL almost no one plays the whole match.

Alt text: Panel a: a falling floor line over a shaded region; football's and rugby union's points sit just above it, the lighter Gaelic points a little higher, a star on the floor, rugby league's points higher to the right and the AFL's near zero. Panel b: six bars between two markers, long for football and rugby league, short near zero for the AFL.

team's stoppages as equation (4) describes, and give each change its stoppage's minute. On the held-out matches, the censored normal gave 4.9 per cent of teams the wrong number of changes, against 13 per cent for a Poisson count. The model put 37 per cent of consecutive changes in one minute, against 6 per cent for minutes picked at random and 33 per cent in the held-out records. For each change on its own, the model and the line miss alike, by 7.7 and 7.8 minutes (Figure 4d). So the model's gain is in how it groups changes. Copying whole sequences from real matches did better still, in readings made after the test. The Electronic Companion's tables let a simulator run the model without those real matches.

What the paper does not do. The paper makes no forecast of how teams would respond to a new limit and draws no link to results, because both need study designs this paper does not have.

6.2 Limitations

Eight limitations apply. First, the agreement between the two rates has a boundary. Baseball is strictly one-way, yet its gap is 24 per cent, because every relief pitcher after a team's first enters for a pitcher who had come on as a substitute. So a small gap needs few chains as well as a ban on re-entry.

Second, the tolerance of 3 per cent was set on football after football's gaps had been read. So the gaps are descriptions, not tests. The predictions for the other sports were entered in the register only after their tests had been written. So the dates cannot show that they came before the data.

Third, rugby union's gap includes temporary replacements, the sin-bin and the front-row replacement rules. The Gaelic panels hold only the finals with complete and consistent lineups.

Fourth, of the 39 predictions in the paper's main register, nine failed and three could not be checked. A ranking of the sports by limit per player failed, because rugby league breaks the ordering. The Electronic Companion places each failure beside its claim.

Fifth, the count models are summaries rather than laws. In rugby union and basketball the censored normal fits worse than a Conway–Maxwell family, and a standard statistical test rejects it in every count panel but football under three substitutions (Electronic Companion).

Sixth, the window describes the middle of the sequence, not its ends. The first change spreads wider than random moments on one window allow, partly because of injuries. Football's last changes pile up at minute 90 because of the source's clock. In rugby union even the middle changes reject one window, and a standard statistical test also rejects football's one width ($p = 0.022$). Every line is fitted and judged on the same matches. Only the model of stoppages was tested on matches it had not seen, and we wrote down its predictions about half an hour before we read those matches. Its pass marks were set after the fitted seasons had been read, and they were looser than its errors there. Only one pass mark set it against a rival method. Its count of changes met its mark narrowly: 4.9 per cent of teams were misplaced, against a mark of 5. The later fitted seasons' own shares misplaced as many. The held-out matches cover one short period under the same rule. So they test the model at another time, not under a new rule. The model puts the last of three or four changes up to 2.4 minutes early. Picking each team's whole sequence of stoppages from real teams removes that error on the fitted seasons. We found this fix only after reading the held-out matches. So four predictions are registered for the 2026–27 seasons, the first under the ten-second exit, whose records we have not read.

Seventh, the comparisons across rules are descriptions, not effects. Football's three- and five-substitution seasons are different years. A

test written down in advance expected the drift across rugby union's eight-bench seasons to be smaller than the jump when the bench grew, and it failed.

Last, the analyses use different samples, and limits are themselves outcomes of rule-making. Football's limit is read from the lineups, so a season whose teams used far less than five could not enter. The AFL, for example, set its first cap against the use it had observed (Australian Football League, 2013).

7 Conclusion

The substitution limit works as a speed limit does. It caps how many starters a team can take off, and the coach decides whom to keep on. In football and rugby league, who plays the whole match depends more on position than on the cap. Where a replaced player cannot return, one line of arithmetic ties the starters withdrawn to the substitutions made. It gives governing bodies a formula and data providers a free check. The arithmetic does not say how teams will respond to a new limit. The number of changes and their timing follow regularities that simulators can use. The model of stoppages met its registered predictions on matches it had not seen.

Data availability

The deposits behind every number in this paper, the software that produces them, and a ledger that maps each printed number to its deposit will be archived with the paper in a public repository. No login, licence or fee was involved, and the deposits hold only aggregates, never a named player. The fits use SciPy (Virtanen et al., 2020) and the figures Matplotlib (Hunter, 2007). The archive identifier is withheld

for anonymized review.

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